Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
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Practical Exercise no. 5

Theme

Feature Engineering

Decision Tree Pruning and Hyperparameter Tuning with GridSearch

Exercise

In machine learning, **feature engineering** is the process of transforming raw data into meaningful features that improve model performance. This provides the algorithm with important information, helping it to detect patterns and make more accurate predictions. Techniques include scaling, encoding and creating new features from existing data.

Decision tree pruning is the process of trimming down a decision tree to reduce its complexity and prevent overfitting. Pruning is applied to prevent overfitting by removing unnecessary branches to make the model simpler and better at generalizing to new data. This improves accuracy, efficiency, and interpretability. It can be achieved by setting a maximum depth or a minimum sample per leaf, or by using cost-complexity pruning.

Hyperparameter tuning with GridSearch is a method of optimizing a model's hyperparameters by systematically testing different combinations within a specified range. This ensures that the model neither underfits nor overfits, maximizing its performance and generalization to new data. Cross-validation is typically used in this process to find the best set of hyperparameters for improved accuracy.

Tasks

The Titanic dataset is intended for use in:

T1. Load the dataset using the *pandas.read_csv(...)* function.

T2. Apply feature engineering and exploratory data analysis procedures.

T3. Define the set of input and output variables for the model.

T4. Prepare and organize the set of case studies from the dataset into training and test data, using the *sklearn.model_selection.train_test_split* function.

T5. Train the decision tree classifier model (*sklearn.tree_DecisionTreeClassifier*) using the training data.

T6. Plot the resulting tree and evaluate the model using the classification report and the confusion matrix.

T7. Tune the model using GridSearchCV *(sklearn.model_selection.GridSearchCV)*. Test hyperparameters such as the *criterion*, *max_depth* and *min_samples_leaf*.

T8. Evaluate the best-developed model and perform the corresponding critical analysis.

T9. Apply the pruning technique to the first model. Try pre- and post-pruning using *max_depth* and *ccp_*alpha respectively.

T10. Repeat **T8** to find the best models resulting from each pruning process.

T11. What conclusions did you draw from the results observed in **T6**, **T8** and **T10**, what conclusions did you draw? In which situations do the model perform better? Which model is the best? Specify the set of parameters. Which model settings should be improved?